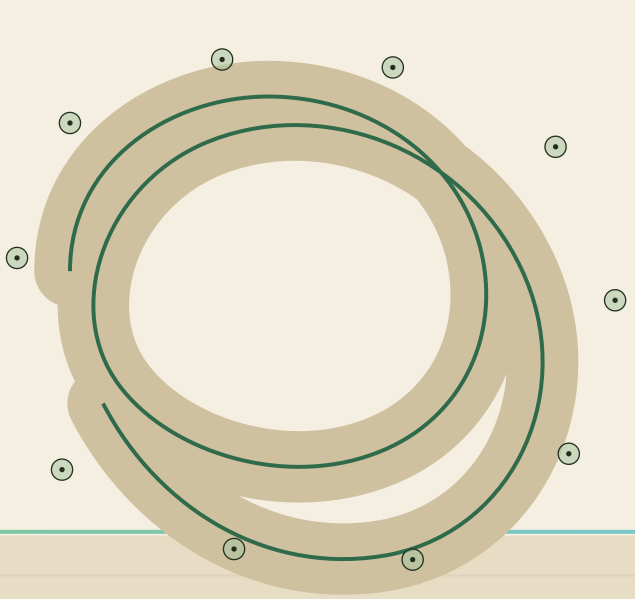


nabta WELLNESS PARK

THE LIVING SHADE LOOP

DUBAI MUNICIPALITY · AI PARK DESIGN CHALLENGE · DRAFT COMPETITION ENTRY

D·01

Executive Summary & Full Presentation Outline

№01 Executive Summary | №14 Full Presentation

«نبتة» NABTA — Wellness Park

Concept: The Living Shade Loop | Deliverable 1 · Executive Summary | Al Safa 2 · 15,000 m² · AED 35M

VISION

> ***A park where comfort makes healthy habits possible.*** *Nabta ("sprout") reimagines Al Safa 2 as a wellness park organized around a continuous 1.0 km shaded loop — part canopy forest, part AI-form-found pergola — comfortable enough to walk, jog and gather from morning school-runs to midnight strolls, eleven months a year. The loop threads ten "garden rooms," so a 15,000 m² neighborhood park behaves like a much larger one: every lap moves the body through plaza, play, fitness, picnic groves, a sensory wadi garden and quiet restoration. AI shaped every decision — from where shadows fall in August to where a seven-year-old runs at 5pm — while our design team made every call.*

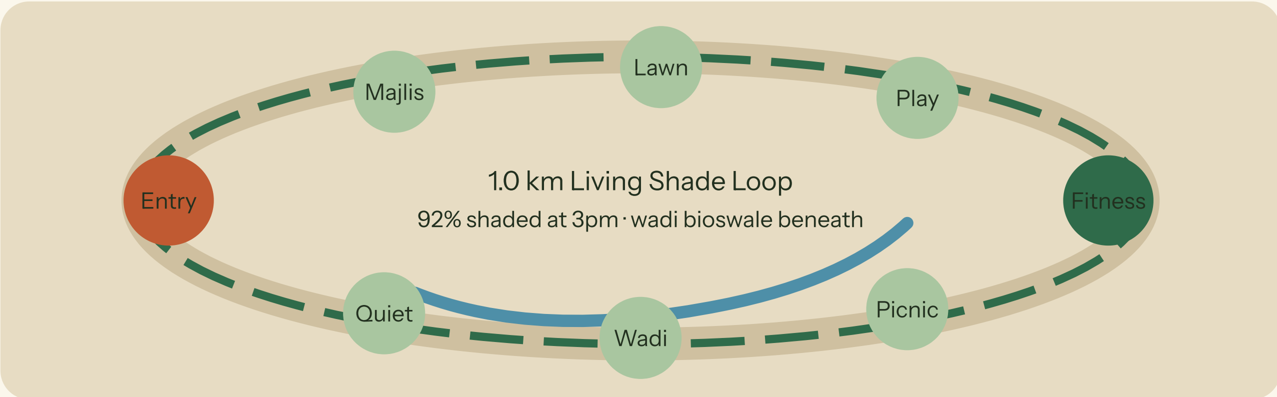
FIVE KEY INNOVATIONS

1. **The Living Shade Loop as health infrastructure.** The 1.0 km walking-jogging track is the park's climatic spine: ghaf, sidr and date-palm canopies alternate with 14 parametric pergola segments form-found against hourly solar simulation — **92% of the loop shaded at 3pm in summer**, cutting perceived temperature by **4.5°C (UTCI)** so daily movement stays safe and habitual.
2. **Agents before architecture.** An agent-based simulation of 1,200 synthetic residents (six AI-assisted personas: toddler families, teens, joggers, seniors, domestic workers, event visitors) stress-tested circulation, desire lines and peak crowding — and projected **~1,800 daily visits** — before a single path was drawn.
3. **Generative masterplanning, human-judged.** 43 masterplan variants were scored on shade continuity, walkability, sightline safety and irrigation demand; the team shortlisted three and hybridized the final scheme in design workshops, documenting every override.
4. **The Wadi Water Loop.** A bioswale garden harvests AC condensate from adjacent buildings plus TSE supply, feeding soil-moisture-driven irrigation of a 100% native palette of **~450 trees** — **41% less irrigation water** than a conventional lawn-based park of equal size.
5. **A park that measures wellbeing.** Anonymous footfall counters, UTCI comfort sensors, adaptive lighting and irrigation telemetry feed a public dashboard reporting active-minutes on the loop, comfort hours per day and water saved — giving Dubai Municipality auditable health and sustainability KPIs, not just maintenance data.

HEADLINE METRICS

- **92%** of the 1.0 km loop shaded at 3pm, summer (vs 31% baseline)
- **−4.5°C** perceived temperature (UTCI) along the loop vs the existing park
- **~1,800** daily visits projected (agent simulation); **ten** garden rooms, play zoned 2–5, 6–12, 13–17
- **~450** native trees (ghaf, sidr, date palm); 78% softscape canopy at year 10

- **41%** reduction in irrigation water vs a conventional lawn-based park
- **100%** of primary routes step-free, tactile-guided, ≤1:21 gradient
- **18-hour** activation day; dawn fitness to evening family strolls under adaptive lighting
- **AED 33.6M** cost plan — 4% contingency inside the AED 35M cap



CONCEPT DIAGRAM: THE SHADED LOOP THREADS THE GARDEN ROOMS; THE WADI BIOSWALE (BLUE) CROSSES THE INTERIOR. SELECTED ROOMS LABELED.

DELIVERABLE 14 — FULL PROJECT PRESENTATION (SLIDE OUTLINE)

#	SLIDE TITLE	CONTENTS
1	NABTA — Wellness Park: The Living Shade Loop	Title, hero aerial render at golden hour, team identity, one-line vision.
2	Al Safa 2 Today	AI site analysis: solar/wind maps, noise from Al Wasl Road, existing tree audit, catchment demographics.
3	Who the Park Serves	Six AI-assisted personas; wellbeing-needs matrix (movement, restoration, social connection); day-in-the-life timelines; ~1,800 daily visits projected.
4	The Big Idea	Shade-first thesis; loop + ten garden rooms diagram; precedent of the sikka and the wadi.
5	AI Workflow Overview	End-to-end pipeline map: analysis → personas → agent simulation → generative options → environmental scoring → human selection.
6	43 Parks, One Winner	Generative masterplan gallery; scoring dashboard (shade, walkability, water); how the team hybridized option 12 + 31.
7	Masterplan	Illustrated 1:500 plan; program areas and m² breakdown; entries, service access, wayfinding logic.
8	Microclimate Proof	UTCI heatmaps before/after; 3pm summer shade animation stills; −4.5°C result and pergola form-finding.
9	Garden Rooms Tour I	Entry plaza, majlis plaza, event lawn, café terrace — renders, capacities, event configurations.
10	Garden Rooms Tour II	Inclusive play dune (by age group), teen zone, fitness grove, picnic groves, sensory wadi, quiet garden; accessibility features throughout.
11	Water & Planting	~450 native trees (ghaf, sidr, date palm, desert grasses); wadi bioswale and condensate loop; 41% irrigation saving math.
12	Measuring Wellbeing	Smart-ops layer: anonymous footfall, comfort sensing, adaptive lighting, irrigation telemetry; public dashboard of active-minutes and comfort hours.

#	SLIDE TITLE	CONTENTS
13	Cost, Phasing, Operations	AED 33.6M elemental cost plan (landscape 42%, pergolas 22%, play/fitness 14%, water systems 12%, smart layer 6%, FF&E 4%) vs 35M cap; 18-month, 3-phase build keeping half the park open; O&M model per DM standards.
14	A Lap of NABTA	60-second AI-visualized walkthrough film; closing metrics strip; call to action.

All AI outputs in this submission were generated, curated and verified under design-team direction; final design decisions are human-made and fully documented in the AI-usage log (Deliverable Package 4).